

AI-Integrated Washing Machine

Mini Project Presentation

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April 11, 2026

Outline

Introduction

Hardware Design

AI & Backend

Mobile App & Testing

Results & Future Scope



SECTION 1

Introduction

1

Problem Statement & Objective

⚠ Problem

Existing semi-automatic machines **waste water and detergent**, and damage clothes due to **fixed, non-adaptive** wash cycles.

🎯 Objective

Convert a semi-automatic machine into a **fully automated, AI-powered** system that:

- Identifies cloth type via smartphone camera
- Optimizes wash parameters per fabric type

★ **Goal:** *Intelligent washing that preserves fabric quality and conserves resources.*

System Architecture



Two-Stage AI Pipeline:

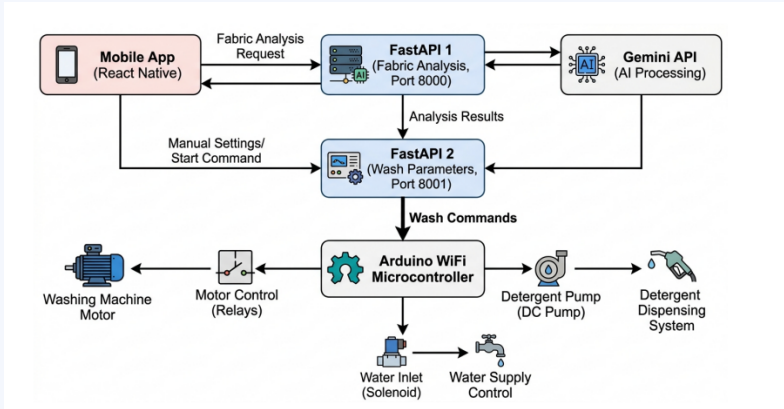
Stage 1: Fabric Detection

Identifies fabric type, fiber category, and soil level.

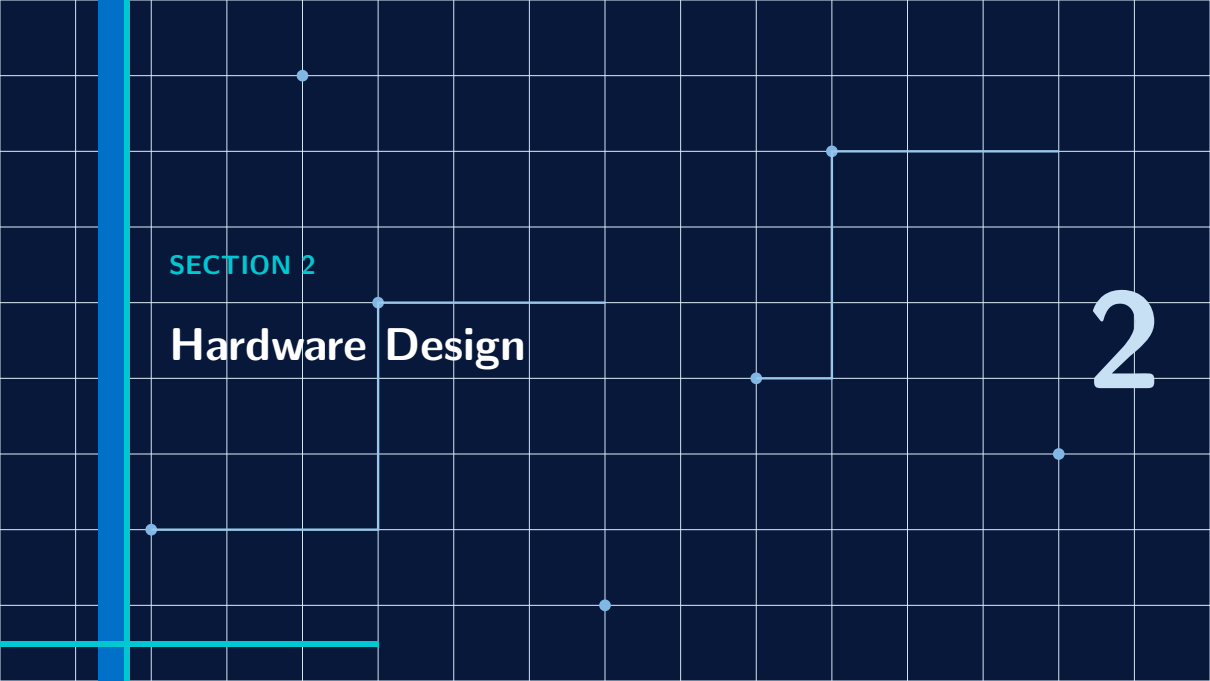
Stage 2: Parameter Calculation

Computes optimal wash settings based on Stage 1 output.

System Architecture Overview



➔ **Flow:** Image → Gemini (Stage 1 & 2) → Arduino → Machine Actuation



SECTION 2

Hardware Design

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Hardware Components

Core & Commutation

- Semi-automatic Washing Machine
- Arduino WiFi R4
- 3-Phase Motor (240V, 10A)
- 2 Rotation Relays + 1 Solenoid Relay

Dispensing & Actuation

- Solenoid Water Inlet Valve
- 5V DC Water Pump (Detergent)
- Transistor TAS122
- 12V L298N DC Motor Driver



All components integrated to automate washing, dispensing, and motor control.

- **Motor Control:** Clockwise 5 s → Stop 2 s → Counter-clockwise 5 s (loop) via relays.
- **Detergent Dispenser:** Linear regression calculates valve open time.
 - Data collected at 50 ms ... 2400 ms intervals.
 - Regression equation implemented in Arduino code.
- **Water Inlet Valve:** Same calibration approach for accurate filling.



Regression Model

$$t = a \cdot V + b$$

t : open time (ms)

V : desired volume (ml)

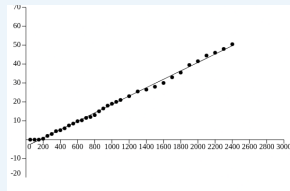
a, b : calibration coefficients

Detergent Dispenser Calibration

Calibration Data

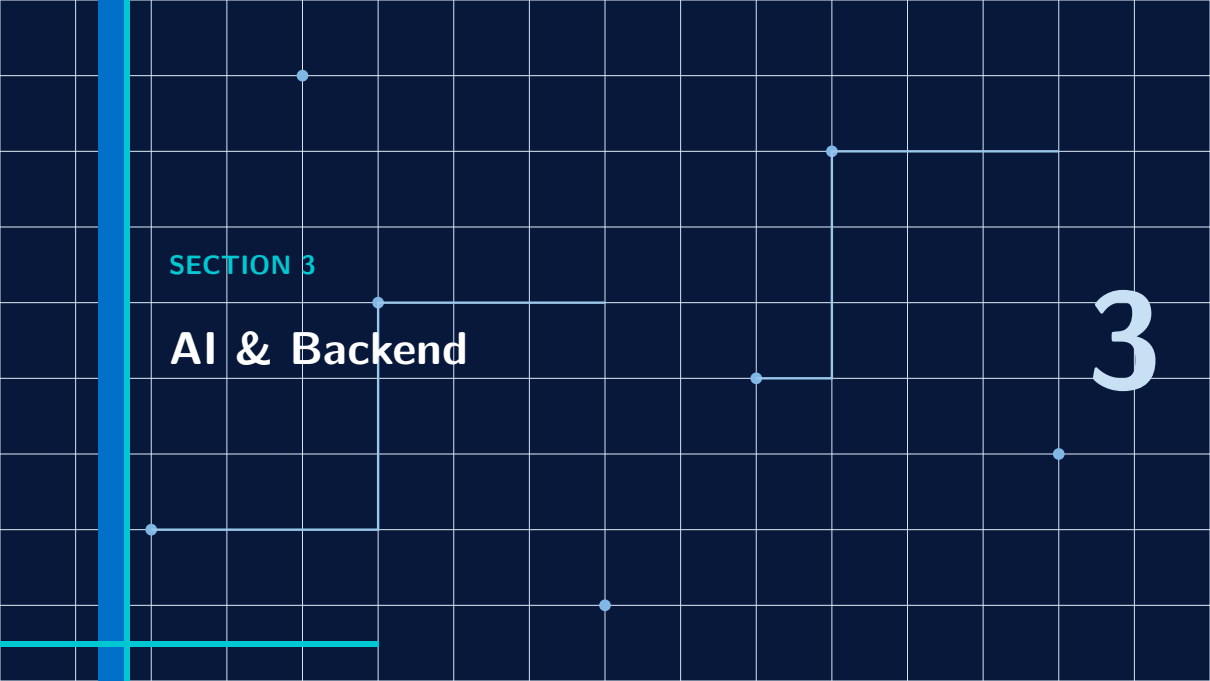
ms	ml	ms	ml	ms	ml
50	0.0	650	10.25	1400	26.5
100	0.0	700	11.5	1500	28.0
150	0.0	750	12.0	1600	30.0
200	0.5	800	13.0	1700	33.0
250	2.0	850	15.0	1800	35.5
300	3.0	900	16.5	1900	39.5
350	4.5	950	18.0	2000	41.5
400	5.0	1000	19.0	2100	44.5
450	6.0	1050	20.0	2200	46.0
500	7.5	1100	21.0	2300	48.0
550	8.5	1200	23.0	2400	50.5
600	9.75	1300	25.5	—	—

Linear Regression



Model Coefficients

Coefficient	Value
a (slope)	0.022176
b (intercept)	2.568697



SECTION 3

AI & Backend

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Multi-Stage Prompt System

Extracts the following from a cloth image:

- Fabric name
- Fiber category
- Soil level (1–5 scale)

Visual markers analyzed:

diagonal ribbing, fiber luster, fold behavior.

Soil Level Scale

- 1 Clean
- 2 Lightly Soiled
- 3 Moderately Dirty
- 4 Heavily Soiled
- 5 Emergency

Stage 1 — JSON Output

 Output JSON | Stage 1: Fabric Analysis

```
{  
  "material_type": "cotton twill",  
  "fiber_category": "Natural",  
  "description": "Heavy cotton fabric with diagonal ribs",  
  "dirt_level": 3,  
  "confidence_score": 0.92  
}
```

Wash Parameter Calculation — Stage 2

Takes JSON from Stage 1 and computes optimal wash settings:

Wash Parameters

- › Detergent: baseline 20 ml, adjusted
- › Soak: 0–30 min
- › Spin: 3–12 min (fabric-based)
- › Water: standard 35 L

Cycle Controls

- › Wash Cycles: 1–2
- › Temperature: up to 60 °C
- › Action: Gentle / Normal / Heavy

Stage 2 — JSON Output

 Output JSON | Stage 2: Wash Parameters

```
{  
  "detergent_ml": 28,  
  "soak_minutes": 15,  
  "spin_minutes": 8,  
  "water_liters": 35,  
  "wash_cycles": 2,  
  "temperature_c": 45,  
  "mechanical_action": "Normal"  
}
```

Two independent microservices:

API 1 — Fabric Analysis (Port 8000)

- Input: Cloth image
- Uses Pillow + Gemini API
- Predicts: fiber type, dirt level, description
- **Human-in-the-Loop** feedback for corrections

API 2 — Wash Parameter Calculator

- Input: Stage 1 JSON
- Outputs optimized wash settings
- Direct serial I/O with Arduino

 **Tech Stack:** Python, FastAPI, Pydantic, Pillow, Gemini API, OpenAPI

SECTION 4

Mobile App & Testing

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Mobile App (React Native Expo)

5 Screens

- 1 Dashboard
- 2 Fabric Analysis
- 3 Wash Progress
- 4 Manual Override
- 5 Settings

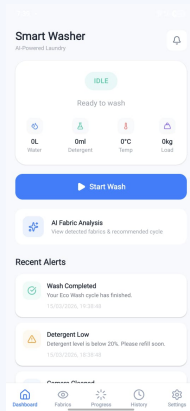
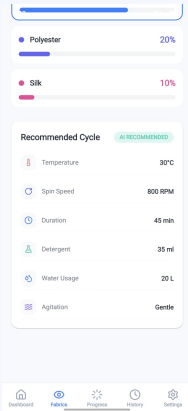
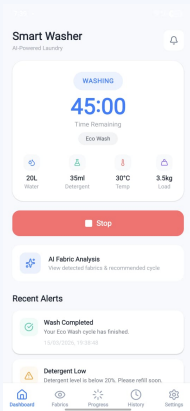
★ Key Features

- Reusable UI components (Step Indicators, Cards)
- TypeScript interfaces for data integrity (*Fabric*, *MachineMetrics*, *CycleState*)
- Tailwind CSS utility-first styling
- Async API service layer connecting to backend
- Cross-platform (iOS & Android)

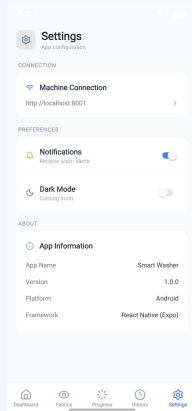
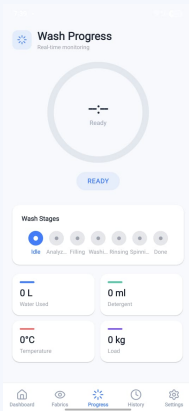
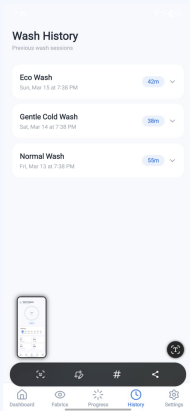


App communicates with FastAPI backends via REST calls.

Mobile App Screenshots (1/2)



Mobile App Screenshots (2/2)



Dataset Validation & Testing

- › Test cases from **heavy-duty (Denim)** to **delicate (Chiffon)**.
- › **Human-in-the-Loop (HITL)** testing for misclassifications:
 - Example: Differentiating *Cotton Twill* vs *Synthetic Blends*
 - Iterative prompt refinement eliminated identification errors.
- › Standardized JSON output ensures seamless hardware integration.
- › Detergent dispensing validated via linear regression (error < 5%).

✓ Result

Reliable classification across **15+ fabric types** with high confidence scores.



SECTION 5

Results & Future Scope

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Results & Key Achievements

- **Accurate Fabric Classification:** High confidence scores across multiple fabric types.
- **Precise Detergent Dispensing:** Linear regression validated against real-world measurements.
- **End-to-End Automation:** Image scan → AI decision → Hardware execution.
- **Increased Cloth Lifetime:** Optimized parameters per fabric type reduce wear and tear.
- **Fully Functional Dual-API System:** Real-time feedback loop for continuous improvement.

Success

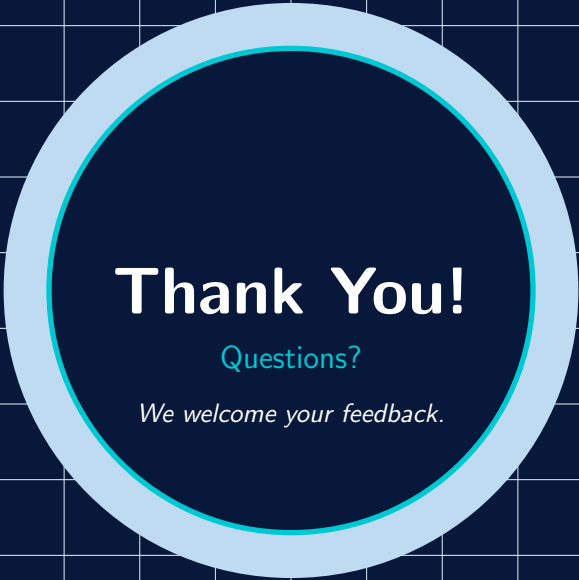
Successfully converted a semi-automatic machine into an **intelligent, adaptive** washing system.

Conclusion

- Demonstrated a **low-cost** AI-integrated laundry automation system.
- Bridges computer vision, prompt engineering, embedded systems, and mobile development.

Future Scope

- Weight sensor for load-based optimization.
- Cloud dashboard for wash history analytics.
- Multi-cloth batch scanning support.
- Smart home integration (Google Home, Alexa).



Thank You!

Questions?

We welcome your feedback.